

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (CCE) (Credit System) Degree Examinations
May - June 2023

20CC602 - NEXT GENERATION TELECOM NETWORKS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the process how the capacity scales linearly with the number of added small cells.	5	L*2	1	1
b)	What is tree scanning? Discuss the beam scanning mechanism for tree scanning?	5	L2	2	1
c)	Illustrate the terminal specific serving cluster with a serving beam from P-SAN blocked.	10	L2	2	1
2. a)	Compare the 5G multiple and medium access schemes for OFDM, PBMC - OQAM, UF- OFDM, SCMA, NOMA and IDMA.	10	L2	1	1
b)	Explain the digital, analog and hybrid beam forming architectures at mm wave technology with diagram. List out their limitations.	10	L2	2	1
3. a)	Explain the role of antennas in mm wave technology.	5	L2	1	1
b)	Elaborate the challenges for radio communication at mm wave bands.	10	L2	2	1
c)	What is system enabler? How it can be utilized for system toolbox?	5	L2	1	1
Unit – II					
4. a)	Explain SDN for 5G systems with diagram and highlight its benefits.	10	L2	3	1
b)	Explain the E- UTRAN architecture with diagram.	10	L2	3	1
5. a)	Explain the relation between functional, logical, orchestration and physical architectures with diagram.	10	L2	3	1
b)	Explain the split functionality between the Local radio access point and the central processor.	10	L2	3	1
6. a)	Describe how the placement functionality may be determined by the type of deployment.	10	L2	3	1
b)	Elaborate the major features of the 4G LTF D2D concept.	10	L2	4	1
Unit – III					
7. a)	Explain the multiuser and single user with massive MIMO base stations using example.	10	L2	5	1
b)	Explain the impact of channel state information errors on the throughput of massive MIMO systems.	10	L2	5	1
8. a)	Explain channel models for massive MIMO.	10	L2	5	1
b)	Explain the beam steering control of antenna blocks with hybrid beam forming.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
